

Use case	Receive Notification	
Release (id)	UC-07, v1.0, 26 April 2026	
Context of Use	The system proactively informs passengers of disruptions affecting their currently active tickets.	
Scope	Allow the system to automatically send real-time email and in-app alerts to passengers regarding route changes or delays that affect their active bookings.	
Level	Subfunction (System-triggered).	
Primary actor	Train Management System.	
StakeHolder & interest	StakeHolder	Interest
	Registered user	Needs prompt information to adjust travel plans.
	Customer service	Benefits from reduced support calls if users are automatically informed.
Preconditions	→ The user has an active ticket/subscription for the affected route and has a valid email address registered.	
Minimal Guarantees	→ The notification is logged in the user's internal dashboard (UC8) even if the external email delivery is delayed.	
Success Guarantees	→ The user receives an email and an in-app alert detailing the specific delay or route change.	
Trigger	The Train Management System registers a significant delay or route change on a scheduled journey.	
Description (Main Success Scenario)	Step	Action
	1	Train Management System detects a delay/route change.
	2	System queries the Ticket Database to identify all users with active bookings for that specific train.
	3	System generates a customized alert message detailing the disruption.
	4	System pushes an in-app notification to the affected users' profiles.
	5	System sends an email notification to the affected users within 1 minute of the event.
Extension	Step	Other action
	5a	The external email service is temporarily down.
	5a1	System queues the emails for automatic retry.
	5a2	The in-app notification is still successfully delivered to the user's dashboard.
Technology & data Variations	Email notifications must strictly comply with the 1-minute delivery constraint (NFR 23).	